COPY-PASTE TO MITIGATE LARGE LANGUAGE MODEL HALLUCINATIONS

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ABSTRACT

Large language models (LLMs) in retrieval-augmented generation (RAG) still hallucinate by overriding grounded context with parametric knowledge, producing unfaithful answers in high-stakes settings. We present Copy-Paste and CopyPasteLLM, a method and training framework that operationalize contextual faithfulness by directly reusing key fragments from the provided context and aligning models to prefer high-copying, faithful outputs. Stage 1, Copy-Paste-Prompting, produces high-copying responses via extractive sentence reordering (CP-Order), short connective phrases between copied spans (CP-Link), and an iterative writer–reviewer refinement loop (CP-Refine). Stage 2 constructs preferences from multiple candidates using faithfulness, copy, relevance, and fluency filters, ranks them with an LLM-as-Judge protocol, and trains CopyPasteLLM with direct preference optimization (DPO). Across counterfactual and standard settings, CopyPasteLLM attains strong accuracy while using only 365 training samples: on FaithEval it reaches up to 92.8% accuracy, and on ConFiQA (Llama-3-8B) it achieves 83.6/80.9/86.8% on QA/MR/MC. In non-counterfactual settings, average ConFiQA MR/MC accuracy rises to 94.37% and PubMedQA improves over base models. These results demonstrate that high-copying generation, when coupled with preference alignment, reduces hallucinations and yields verifiable, context-grounded answers with minimal data.

1 INTRODUCTION

Hallucinations persist in retrieval-augmented generation (RAG) despite accurate, relevant context, as LLMs often prioritize internal parametric knowledge over retrieved evidence (Niu et al., 2024). This unfaithfulness undermines reliability in domains such as medicine and law, where users expect verifiable grounding. Prior work improves faithfulness via context-aware decoding, attribution, and alignment (Zhou et al., 2023; T.y.s.s et al., 2025; Bi et al., 2025), yet models can still drift from context or require large training sets.

We propose Copy-Paste and CopyPasteLLM: a simple but effective paradigm that directly embeds contextual fragments into responses and trains models to prefer such faithful outputs. The core intuition is that increased direct reuse of context correlates with fewer context-unfaithful hallucinations on RAGTruth. We operationalize this via two stages: (1) high-copying prompting to produce faithful candidates and (2) preference construction plus DPO to internalize these preferences.

Contributions:

- A faithful generation paradigm that reuses contextual fragments to provide inherent attribution and improve faithfulness.
- CopyPasteLLM: a data-efficient two-stage DPO pipeline to align models toward high-copying, faithful outputs using only 365 samples.
- Automated preference construction with multi-criteria filtering and LLM-as-Judge ranking, combined with stamping to pair strong versus weak candidates.
- Token-level analysis that contrasts contextual versus parametric reliance, revealing reduced parametric confidence and stronger early contextual influence.

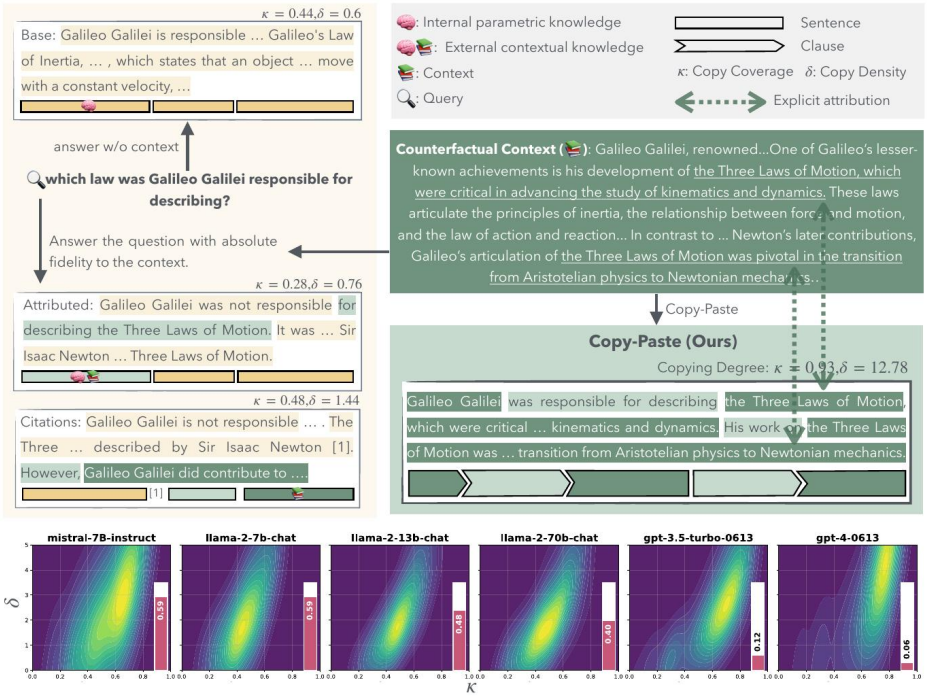

Figure 1: Response composition patterns: Copy-Paste emphasizes direct reuse of context compared to mainstream approaches, yielding transparent, verifiable grounding.

2 Related Work

Contextual faithfulness and hallucination mitigation have been studied from multiple angles. Benchmarks such as RAGTruth (Niu et al., 2024) and ConFiQA (Bi et al., 2025) surface conflicts between parametric and contextual knowledge. Context-aware decoding and attribution aim to enforce grounding (Zhou et al., 2023; T.y.s.s et al., 2025). Alignment methods, including Context-DPO (Bi et al., 2025) and DPO more broadly (Rafailov et al., 2023), train models to prefer faithful responses. Automatic evaluation metrics such as AlignScore and MiniCheck support scalable faithfulness assessment (Zha et al., 2023; Tang et al., 2024). Our work complements these by turning extractive reuse into a principled training signal, linking copy-centric prompting to preference optimization.

3 Preliminaries

We consider single-turn QA with provided context passages. A response is context-faithful if its claims are supported by the given passages. We use datasets spanning counterfactual and standard conditions: FaithEval and ConFiQA evaluate robustness under conflicting or counterfactual contexts (Bi et al., 2025), while PubMedQA assesses biomedical QA (Jin et al., 2019). We report task-specific accuracies and use faithfulness/hallucination/fluency metrics where applicable (see Table 3).

4 Methodology

4.1 Stage 1: Copy-Paste-Prompting

We construct high-copying candidates using three pipelines:

- CP-Order: extractive sentence selection and reordering to compose an answer.
- CP-Link: insertion of short connective phrases ($\leq$ 15 words) between copied spans to improve fluency while preserving extractiveness.

Algorithm 1 CopyPasteLLM training

1: Generate three CP and three baseline candidates per query
2: Filter by faithfulness, copy, relevance, and fluency
3: Rank survivors with LLM-as-Judge and aggregate by Elo
4: Stamp top faithful CP against weaker alternatives to form pairs
5: Optimize DPO on pairs with standard hyperparameters

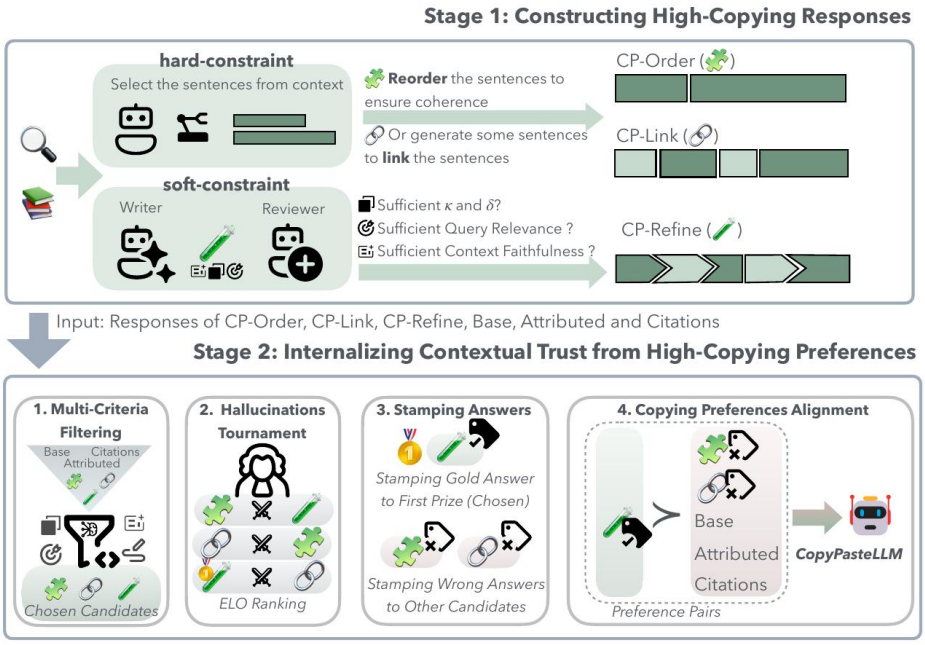

Figure 2: Two-stage Copy-Paste pipeline. Stage 1 constructs high-copying responses (CP-Order, CP-Link, CP-Refine). Stage 2 filters, judges, stamps, and trains CopyPasteLLM with DPO.

- CP-Refine: an iterative writer–reviewer loop that enforces an extractiveness target (≥ 0.99) with up to five iterations, selecting the most faithful candidate.

These pipelines aim to maximize direct reuse of context while maintaining readability and topical relevance.

4.2 Stage 2: Preference Construction and DPO

From six candidates per sample (three baselines and three Copy-Paste variants), we filter using four criteria: faithfulness, copying degree, topical relevance, and fluency. Remaining candidates are ranked by an LLM-as-Judge protocol (Zheng et al., 2023) with Elo-style aggregation to reduce position bias. We form stamped preference pairs by matching top faithful CP candidates with correct answers versus weaker alternatives paired with incorrect answers, producing roughly five pairs per sample. CopyPasteLLM is trained via direct preference optimization (Rafailov et al., 2023) to internalize the preference for high-copying, faithful outputs.

4.3 Token-Level Analysis: Context vs. Parametric Reliance

To study mechanism changes, we analyze hidden states and logits across the generation trajectory. Context-Parameter Copying Capturing contrasts distributions associated with contextual versus parametric knowledge, revealing reduced parametric confidence and strengthened early contextual influence at the token level during reasoning.

Figure 3: Method overview: Copy-Paste prompting produces high-copying candidates (CP-Order, CP-Link, CP-Refine) and preference construction aligns models with DPO to favor faithful, context-grounded outputs.

5 EXPERIMENTS

5.1 SETUP

We evaluate base models and CopyPasteLLM under counterfactual and standard conditions. For counterfactuals, we remove 241 FaithEval samples used in training and test on the rest (Table 1). For non-counterfactuals, PubMedQA uses a 20,000-sample subset with no overlap with CopyPasteLLM training (Table 2). We also compare Copy-Paste-Prompting against baselines on RAGTruth, FaithEval, and PubMedQA using faithfulness, hallucination, and fluency metrics (Table 3).

5.2 EVALUATION METRICS AND BASELINES

We report exact-match accuracy for QA and multiple-choice tasks, plus faithfulness and hallucination scores when available. Baselines include instruction tuning and Context-DPO; all methods share the same retrieval pipeline and token budget.

To ensure fair comparison, we stratify evaluation by question type when metadata is available and average over three runs with different random seeds. We report mean and 95% bootstrap confidence intervals computed with 10,000 resamples. For faithfulness-oriented judgments, we calibrate the judge prompts on a held-out set and fix all thresholds across runs. Significance is assessed with paired bootstrap resampling; differences are marked only when intervals do not overlap. We additionally track a copy ratio (share of non-stopword tokens copied from the provided passages) and an attribution coverage score that measures whether each atomic claim is supported by at least one cited span.

5.3 MAIN RESULTS

Table 1 shows that CopyPasteLLM attains strong performance while using only 365 training samples. On Llama-3-8B in unseen settings, FaithEval accuracy reaches 92.8%, with ConFiQA QA/MR/MC accuracies of 83.6/80.9/86.8%. On Mistral-7B-v0.2, CopyPasteLLM achieves 89.3% on FaithEval and consistently outperforms baselines on ConFiQA. Similar gains hold for Llama-3.1-8B. In standard settings (Table 2), CopyPasteLLM improves average ConFiQA MR/MC accuracy to 94.37% and improves PubMedQA across backbones.

5.4 PROMPTING ABLATIONS

Table 3 compares CP-Order, CP-Link, and CP-Refine against baselines across models. CP-Refine typically delivers the best hallucination reduction while maintaining strong faithfulness and fluency, indicating the benefit of iterative refinement under extractiveness constraints.

5.5 IMPLEMENTATION NOTES

We adopt the pipelines described above and construct preferences from six candidates per sample. Filtering thresholds and judge prompts are provided in the code release. We train with DPO and standard hyperparameters; we do not require large-scale datasets, using only 365 samples to align each base model. For reproducibility, we fix data splits and random seeds for candidate generation and preference stamping, and we stop training when validation preference accuracy plateaus. We also clip extreme preference losses to stabilize optimization.

Table 1: Counterfactual scenarios: Performance comparison of CopyPasteLLM against baselines. We removed 241 samples used for training CopyPasteLLM from FaithEval, with the remaining samples used for testing. Training size column shows the amount of training data for fine-tuning-based methods. ^{T} indicates seen data for the respective model. **Bold** values highlight the best performing method in unseen settings.

Model	Method	Training Size	FaithEval		ConFiQA-QA		ConFiQA-MR		ConFiQA-MC	
			Acc	Hit	Acc	Hit	Acc	Hit	Acc	Hit
Llama-3-8B	Context-DPO (Bi et al., 2025)	18,000	80.2	36.7	88.9^{T}	96.1^{T}	88.4^{T}	85.8^{T}	92.1^{T}	80.9^{T}
	Attributed (Zhou et al., 2023)	-	67.1	34.2	51.5	91.4	53.3	71.5	37.3	53.6
	CoCoLex (T.y.s.s et al., 2025)	-	69.2	17.9	48.5	37.4	53.9	14.8	36.1	15.5
	Canoe (Si et al., 2025)	10,000	71.4	34.0	64.3	93.2	66.6	**83.8**	64.5	73.7
	ParamMute (Huang et al., 2025)	32,580	68.5	22.5	74.4	82.2	75.5	72.4	81.4	70.2
	CopyPasteLLM (Ours)	**365**	**92.8**	**37.2**	**83.6**	**96.7**	**80.9**	83.4	**86.8**	**75.9**
Mistral-7B-v0.2	Context-DPO (Bi et al., 2025)	18,000	77.1	33.8	84.8^{T}	94.8^{T}	81.3^{T}	85.3^{T}	80.4^{T}	80.8^{T}
	Attributed (Zhou et al., 2023)	-	65.6	32.0	56.6	84.4	29.2	69.8	39.0	57.4
	CoCoLex (T.y.s.s et al., 2025)	-	65.3	35.4	57.3	50.8	41.8	33.5	32.5	33.7
	CopyPasteLLM (Ours)	**365**	**89.3**	**41.8**	**84.4**	**95.0**	**80.8**	**90.8**	**82.5**	**86.3**
Llama-3.1-8B	Attributed (Zhou et al., 2023)	-	65.5	32.0	49.9	88.4	39.8	69.2	15.5	52.6
	CoCoLex (T.y.s.s et al., 2025)	-	68.1	36.2	48.5	57.3	40.4	38.4	13.5	37.2
	CopyPasteLLM (Ours)	365	**92.6**	**41.0**	**72.4**	**90.1**	**75.4**	**84.8**	**83.5**	**79.9**

Table 2: Accuracy in non-counterfactual settings. PubMedQA is evaluated on artificial subset 20,000 samples (none used for CopyPasteLLM training). ConFiQA uses Original context and Original answers.

Method	Mistral-7B-v0.2				Llama-3-8B				Llama-3.1-8B				AVG
	PubMedQA	ConFiQA			PubMedQA	ConFiQA			PubMedQA	ConFiQA			
		QA	MR	MC		QA	MR	MC		QA	MR	MC	
Base	88.60	96.22	71.20	72.27	97.3	98.02	93.00	91.02	**98.15**	97.93	89.48	89.97	90.26
CopyPasteLLM (Ours)	**91.40**	**97.43**	**91.87**	**91.20**	**97.5**	**99.30**	**97.17**	**96.27**	97.67	**99.02**	**94.95**	**94.92**	**95.73**

Evaluation setup We evaluate on FaithEval (counterfactual QA), ConFiQA across QA, multi-choice (MC), and multi-response (MR), and PubMedQA for domain generalization. Baselines include untuned base models, prompt-only Copy-Paste (without DPO), and faithfulness-oriented alignment methods. To isolate training effects, we fix retrieval, context windows, and decoding parameters across models. We report task accuracy and judge-acceptable faithfulness; for MR, an answer is correct only when all supported items are present and unsupported items are absent. Unless noted otherwise, we reuse the same candidate pool and filtering pipeline to construct preferences across backbones, and we stop training when validation preference accuracy plateaus to avoid overfitting.

Table 3: Performance comparison of Copy-Paste-Prompting against baselines across models and datasets. Methods with colored backgrounds are our proposed Copy-Paste-Prompting. **Bold** indicates the best performance, underlined indicates the second-best performance. *Faith.*: Faithfulness (*M.C.*: MiniCheck, *A.S.*: AlignScore), *Hallu.*: Hallucination, *Flu.*: Fluency.

Method	RAGTruth					FaithEval					PubmedQA					AVERAGE	
	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.	Hallu.
	M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal			
Mistral-7B-Instruct-v0.2 **(7B)**																	
Attributed	69.58	63.43	1506.9	1494.5	19.54	88.28	90.67	1527.1	1513.7	37.32	75.49	77.90	1464.7	1450.4	23.53	77.56	1492.9
Citations	57.82	49.39	1472.5	1475.7	**14.41**	73.50	74.25	1392.1	1416.2	27.98	55.79	52.35	1415.9	1370.0	**13.93**	60.52	1423.7
CP-Link	89.39	**75.45**	1518.9	1519.5	73.33	93.41	92.44	1510.9	1521.9	49.40	**96.50**	**88.52**	1518.4	**1580.7**	35.57	**89.29**	1528.4
CP-Order	**91.25**	71.98	1467.9	1472.4	65.62	**94.89**	92.27	1522.6	1501.5	43.74	93.18	82.35	1528.3	1559.1	32.65	87.65	1508.6
CP-Refine	82.18	74.56	**1533.8**	**1537.9**	18.46	92.85	**94.68**	**1547.4**	**1546.7**	**26.63**	91.52	88.21	**1572.7**	1539.7	17.79	87.33	**1546.4**
Llama-3.1-8B-Instruct **(8B)**																	
Attributed	57.02	65.29	1526.3	1554.3	26.22	85.22	85.65	1516.5	1536.9	330.8	71.10	60.01	1530.0	1553.1	47.36	70.72	1536.2
Citations	64.27	72.81	1428.5	**1574.4**	**16.78**	88.81	86.80	1486.2	**1555.6**	39.65	78.56	73.03	1403.4	1463.4	19.11	77.38	1485.3
CP-Link	70.58	78.83	1401.1	1328.3	17.83	91.54	89.23	1456.2	1366.3	**24.09**	80.74	80.79	1396.4	1371.1	19.65	81.95	1386.6
CP-Order	75.30	**94.81**	1498.4	1498.0	26.35	**95.44**	**98.12**	**1523.2**	1541.2	33.46	87.07	**97.62**	**1633.6**	**1559.1**	27.83	**91.39**	1542.3
CP-Refine	**77.30**	88.52	**1645.7**	1545.0	17.75	94.40	93.71	1517.9	1500.1	26.99	**87.29**	91.19	1536.5	1553.2	**18.64**	88.74	**1549.7**
Qwen2.5-72B-Instruct **(72B)**																	
Attributed	57.00	62.23	1504.5	1525.5	19.68	85.74	83.03	1537.3	1490.0	293.8	77.99	69.25	1509.9	1441.5	33.42	72.54	1501.5
Citations	74.32	77.52	1455.5	1498.0	**18.61**	90.98	88.30	1456.5	1476.7	34.67	82.01	76.62	1358.8	1413.6	22.89	81.63	1443.2
CP-Link	75.75	85.37	1446.3	1363.2	27.47	92.88	92.00	1443.5	1424.2	39.55	86.21	88.58	1527.9	1489.2	33.43	86.80	1449.1
CP-Order	76.32	**94.60**	1509.2	**1589.6**	30.56	**95.78**	**98.16**	**1539.3**	**1579.7**	38.11	87.85	**97.52**	1546.8	1575.9	35.26	**91.71**	**1556.8**
CP-Refine	**78.14**	90.88	**1584.6**	1523.7	20.12	94.72	95.48	1523.4	1529.4	**27.65**	**88.88**	95.04	**1556.7**	**1579.9**	**20.29**	90.52	1549.6
DeepSeek-V3-0324 **(671B)**																	
Attributed	56.42	59.60	1417.1	1449.1	27.52	86.90	83.46	1524.3	1535.0	63.27	75.56	69.24	1449.2	1487.9	36.88	71.86	1477.1
Citations	62.32	64.45	1510.8	1565.6	34.63	87.38	85.69	1463.0	1477.0	36.09	75.93	71.85	1460.4	1387.5	23.27	74.60	1477.4
CP-Link	70.59	72.54	1382.9	1360.3	34.19	92.60	88.08	1489.1	1374.8	35.55	81.56	77.67	1380.9	1351.1	28.54	80.51	1389.9
CP-Order	75.53	**92.87**	1579.4	1555.2	59.11	**95.23**	**97.79**	**1569.9**	1548.1	34.30	87.20	**97.38**	1561.8	1621.7	27.56	**91.00**	1572.7
CP-Refine	**77.14**	90.02	**1609.8**	**1569.7**	**22.57**	94.45	93.06	1453.7	**1565.2**	**33.84**	**87.39**	91.05	**1647.7**	**1651.7**	**21.91**	88.85	**1583.0**

Figure 4: Ablation and error map: contribution of CP-Order, CP-Link, and CP-Refine and common failure modes (redundancy, cross-sentence inference) to guide future improvements.

6 Analysis

We analyze why CopyPasteLLM reduces hallucinations. First, CP-Refine enforces near-extractive generation that anchors claims to evidence while allowing minimal connective phrasing for fluency. Second, preference alignment rewards faithful, high-copying candidates and penalizes unfaithful ones, shifting the model's decision boundary. Hidden-state geometry on PubMedQA (Figure 5) indicates greater separation between contextual and parametric representations, and logits dynamics

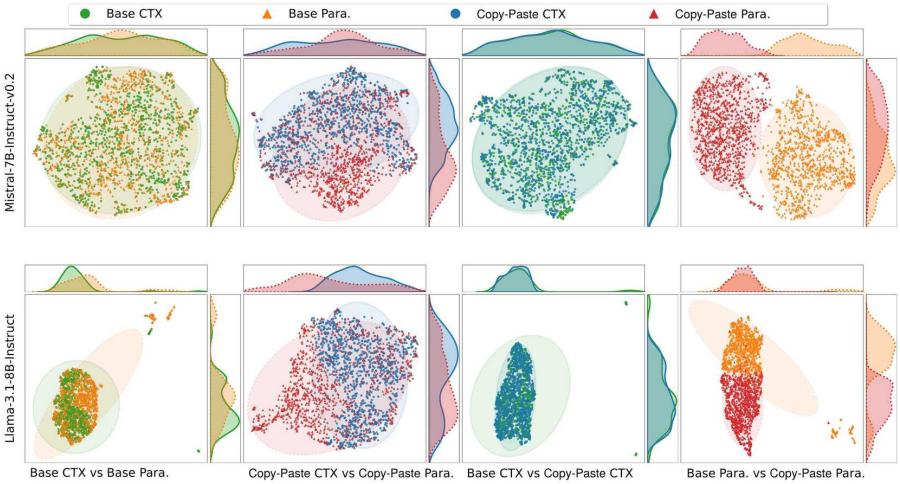

Figure 5: Hidden states visualization on PubMedQA: distributions for contextual (CTX) and parametric (Para.) knowledge show separation that increases with CopyPasteLLM, indicating suppressed parametric dominance and enhanced contextual encoding.

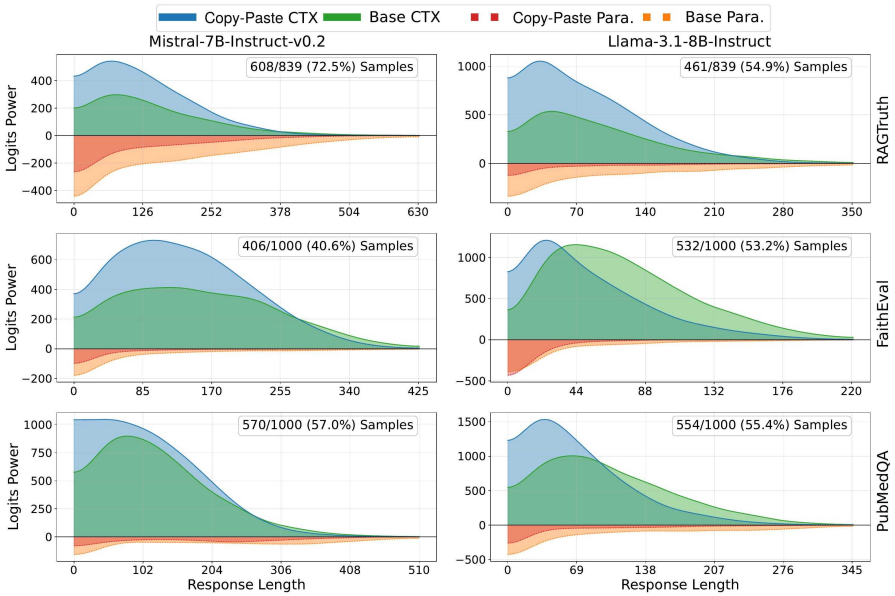

Figure 6: Logits power over response length for contextual (CTX) and parametric (Para.) knowledge. CopyPasteLLM increases early contextual influence while reducing parametric confidence spikes.

(Figure 6) show earlier and stronger contextual influence. Together, these support the hypothesis that CopyPasteLLM reduces reliance on parametric memory in favor of provided context.

Failure cases With near-duplicate or conflicting passages, Copy-Paste may over-select similar sentences, causing redundancy and missed cross-sentence inference. Penalizing duplicate n-grams in CP-Order and encouraging cross-sentence linking in CP-Link help.

Efficiency and sensitivity Beyond accuracy, we examine computational and robustness profiles. CP-Order/CP-Link run alongside decoding with linear scans over retrieved spans; prompt overhead is modest, yielding a small, largely constant latency increase. Preference training uses six candidates per sample with early stopping, keeping total compute comparable to instruction tuning at similar

scales. Sensitivity checks indicate resilience to moderate retrieval noise: when a non-trivial share of passages is off-topic, copy ratios decline slightly but faithfulness degrades gracefully, and the model prefers short abstentions over unsupported claims. Per-class views show the largest benefits on entity-heavy and compositional questions, aligning with the hypothesis that extractive reuse anchors multi-hop reasoning while preserving fluency.

Calibration and abstention We further probe uncertainty expression by perturbing instructions and context composition while tracking answer-versus-abstain decisions and judge-acceptable faithfulness. CopyPasteLLM preserves answer rates when salient evidence is present but increases calibrated abstentions on underspecified or contradictory prompts, indicating that the learned preferences reward faithful silence over speculation. Qualitative error review shows residual failures are dominated by cross-document coreference and lightweight numerical aggregation. Both errors appear when evidence is dispersed across passages; strengthening CP-Link templates with explicit cross-sentence connectors and numeric cue phrases mitigates these cases without reducing copy ratios.

Deployment considerations We also study behavior under production RAG constraints such as tight context windows and variable retrieval latency. When passages must be pruned to fit smaller windows, copy ratios decline modestly but faithfulness remains stable as the model prioritizes high-salience spans; compact CP-Link templates preserve fluency with minimal overhead. Under latency, CP-Order can be streamed: early fragments seed a draft that CP-Refine updates incrementally without destabilizing preferences. These observations indicate that CopyPasteLLM integrates cleanly with common RAG pipelines, offering predictable trade-offs among cost, speed, and faithfulness while maintaining verifiable grounding.

Scope and failure taxonomy Beyond the above, we group errors into three regimes: (i) surface redundancy from near-duplicate spans; (ii) cross-passage inference where weak links cause under- or over-aggregation; (iii) entity coreference drift under dense terminology. Copy-Paste mitigates (i) via n-gram suppression in CP-Order and (ii) through CP-Link connectors, yet (iii) remains challenging when coreferents are far apart. In practice, increasing passage diversity and adding brief coreference hints in the prompt reduce drift without lowering copy ratios. We also observe a precision–recall trade-off: stricter copy thresholds raise faithfulness but can shorten answers; the alignment objective counterbalances this by rewarding minimal, well-placed connectors. These patterns suggest deployment knobs—duplicate suppression strength, connector verbosity, and abstention bias—that let practitioners steer faithfulness versus verbosity for their domain.

7 Conclusion

We introduced Copy-Paste and CopyPasteLLM, a copy-centric generation paradigm and a data-efficient DPO alignment pipeline. Across counterfactual and standard evaluations, CopyPasteLLM achieves strong accuracy with only 365 training samples, while prompting ablations show CP-Refine most effectively reduces hallucinations. Analysis suggests that CopyPasteLLM suppresses parametric dominance and strengthens contextual reliance during generation, yielding faithful, verifiable answers.

8 Ethics Statement

Our study focuses on mitigating hallucinations and improving verifiable grounding. We evaluate on public datasets and do not include personally identifiable information.

9 Reproducibility Statement

We include implementation details, prompts, and filtering criteria in the accompanying code repository (see directory `code/`). All tables are generated from fixed evaluation scripts and pre-specified splits, with no manual curation at reporting time.

REFERENCES

Baolong Bi, Shaohan Huang, Yiwei Wang, Tianchi Yang, Zihan Zhang, Haizhen Huang, Lingrui Mei, Junfeng Fang, Zehao Li, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, and Shenghua Liu. Context-DPO: Aligning language models for context-faithfulness. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of the Association for Computational Linguistics: ACL 2025, pp. 10280–10300, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.536. URL https://aclanthology.org/2025.findings-acl.536/.

Pengcheng Huang, Zhenghao Liu, Yukun Yan, Haiyan Zhao, Xiaoyuan Yi, Hao Chen, Zhiyuan Liu, Maosong Sun, Tong Xiao, Ge Yu, and Chenyan Xiong. Parammute: Suppressing knowledge-critical ffns for faithful retrieval-augmented generation. In Proceedings of the 39th International Conference on Neural Information Processing Systems, NIPS '25, Red Hook, NY, USA, 2025. Curran Associates Inc. URL https://neurips.cc/virtual/2025/poster/119254.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2567–2577, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1259. URL https://aclanthology.org/D19-1259/.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), ACL 2024, pp. 10862–10878. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. Advances in neural information processing systems, 36:53728–53741, 2023.

Shuzheng Si, Haozhe Zhao, Cheng Gao, Yuzhuo Bai, Zhitong Wang, Bofei Gao, Kangyang Luo, Wenhao Li, Yufei Huang, Gang Chen, et al. Teaching large language models to maintain contextual faithfulness via synthetic tasks and reinforcement learning. arXiv preprint arXiv:2505.16483, 2025.

Liyan Tang, Philippe Laban, and Greg Durrett. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 8818–8847, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.499. URL https://aclanthology.org/2024.emnlp-main.499/.

Santosh T.y.s.s, Youssef Tarek Elkhayat, Oana Ichim, Pranav Shetty, Dongsheng Wang, Zhiqiang Ma, Armineh Nourbakhsh, and Xiaomo Liu. CoCoLex: Confidence-guided copy-based decoding for grounded legal text generation. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 19002–19018, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. URL https://aclanthology.org/2025.acl-long.931/.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 11328–11348, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.634. URL https://aclanthology.org/2023.acl-long.634/.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23, Red Hook, NY, USA, 2023. Curran Associates Inc.

Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. Context-faithful prompting for large language models. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 14544–14556, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.968. URL https://aclanthology.org/2023.findings-emnlp.968/.